

Computer Graphics

CSIT — 3rd Semester — 2082 — Answer Sheet
Subject Code: CSC214

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

- 1. Explain Bresenham's line drawing algorithms. Compare it with DDA line algorithm. Draw a line from (2, 3) to (10, 8) by using Bresenham's algorithm.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 2. What is a polygon mesh? Describe its types. Construct a polygon table and edge table for a cube of side 2 units placed with one vertex at origin.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 3. Explain the Painter's algorithm for visible surface detection. Explain the BSP tree method used for visible surface determination. How does it divide the space and organize objects in a scene?**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

- 4. Explain the need for machine-independent graphics languages. How do such standards benefit application developers?**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 5. Define 2D rotation in computer graphics. Derive the rotation matrix and calculate the new coordinates of a point (2, 3) after a rotation of 45° about the origin.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

- 6. How is the transformation matrix computed when switching from one coordinate system to another? Illustrate with an example.**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain the 3D viewing pipeline in computer graphics. Explain about how a 3D world coordinate system is transformed to a 2D screen?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. For control points $P_0(0,0)$, $P_1(1,2)$, $P_2(3,3)$, and $P_3(4,0)$, calculate the Bezier curve point at $u = 0.5$. Also plot the rough curve shape.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is spatial-partitioning representation? Explain how it differs from boundary representation in terms of geometry storage and processing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is constant intensity shading? Compare Phong shading and fast Phong shading.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Discuss the use of virtual reality in education. How does VR enhance student engagement and learning outcomes?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on a. Lighting in OpenGL. b. Orthographic projection.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.